GRIGORIOS FLOUDAS

MECHANICAL ENGINEERING STUDENT
AT ARISTOTLE UNIVERISTY OF THESSALONIKI

Computational Modeling of Laminar Flow in a Rectangular Cavity

Table of Contents

1. Introduction 3

2. Discretized Relations 4

2.1. Explicit Method 4

2.2. Implicit Method 6

2.3. Boundary Conditions 9

3. Results 11

3.1. Re=1 11

3.2. Re=100 13

3.3. Re=500 15

4. Grid Independence Study 18

5. Opposite Direction of Flow 21

1. Introduction

The objective of the present study is the investigation of incompressible flow of a real fluid inside a rectangular cavity. The cavity consists of three stationary walls, the bottom, the left, and the right, and one moving wall (the top) which moves with a velocity U. The top wall has continuous velocity (assumed to have infinite length) and, as it moves, it drags the fluid due to its contact (because of the fluid's viscosity) and stirs it. This stirring generates vortices inside the cavity, depending on the Reynolds number, $\mathrm{Re} = \mathrm{UL}/\nu$, where L is the horizontal length of the cavity. The following figure illustrates a schematic of the problem.

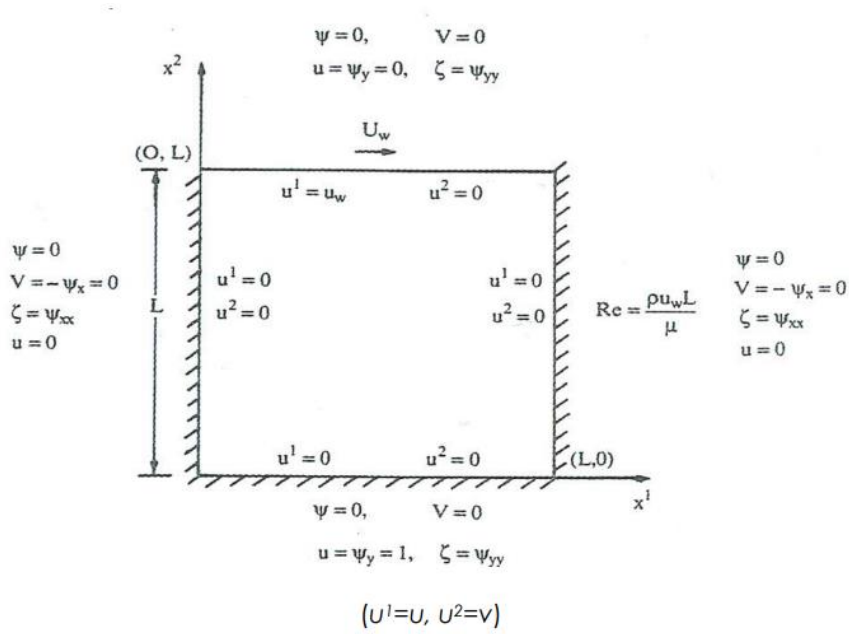

Figure 1.1. *Problem sketch*

The problem is described by the vorticity transport equation:

$$\frac{\partial \zeta}{\partial t} + u\frac{\partial \zeta}{\partial x} + v\frac{\partial \zeta}{\partial y} = \frac{1}{\mathrm{Re}}\left(\frac{\partial^2 \zeta}{\partial x^2} + \frac{\partial^2 \zeta}{\partial y^2}\right) \tag{1.1}$$

which can also be expressed using the stream function as

$$\frac{\partial \zeta}{\partial t} = -\frac{\partial \Psi}{\partial y}\frac{\partial \zeta}{\partial x} + \frac{\partial \Psi}{\partial x}\frac{\partial \zeta}{\partial y} + \frac{1}{\mathrm{Re}}\left(\frac{\partial^2 \zeta}{\partial x^2} + \frac{\partial^2 \zeta}{\partial y^2}\right) \tag{1.2}$$

together with the relationship between the stream function and vorticity

$$\frac{\partial^2 \Psi}{\partial x^2} + \frac{\partial^2 \Psi}{\partial y^2} = -\zeta \tag{1.3}$$

2. Discretized Relations

2.1. Explicit Method

In Computational Fluid Dynamics (CFD), an explicit method refers to a numerical scheme for solving the governing equations of fluid flow, where the solution is computed directly in time. In explicit methods, the solution at the next timestep is calculated directly from known information at the current step, without solving a system of equations.

In terms of time integration, explicit methods are contrasted with implicit methods. In an explicit method, the solution for the unknown variables is obtained straightforwardly, often through a finite difference approach, without solving algebraic systems at each time step. However, explicit methods may have stability limitations that restrict the time step size.

The basic idea is to separate the spatial and temporal domains and then update the solution at each grid point step-by-step using information from neighboring points. The direct nature of the method allows simplicity in implementation but may come at the cost of stability restrictions.

Initially, for the vorticity transport equation, the derivative terms can be expressed using central finite differences as

$$\frac{\zeta_{i,j}^{n+1} - \zeta_{i,j}^{n}}{\Delta t} = -\frac{\Psi_{i,j+1}^{n} - \Psi_{i,j-1}^{n}}{2\Delta y}\frac{\zeta_{i+1,j}^{n} - \zeta_{i-1,j}^{n}}{2\Delta x} + \frac{\Psi_{i+1,j}^{n} - \Psi_{i-1,j}^{n}}{2\Delta x}\frac{\zeta_{i,j+1}^{n} - \zeta_{i,j-1}^{n}}{2\Delta y} + \frac{1}{Re}\left(\frac{\zeta_{i+1,j}^{n} - 2\zeta_{i,j}^{n} + \zeta_{i-1,j}^{n}}{\Delta x^2} + \frac{\zeta_{i,j+1}^{n} - 2\zeta_{i,j}^{n} + \zeta_{i,j-1}^{n}}{\Delta y^2}\right) \tag{2.1.1}$$

To compute the vorticity at the next time step $(n+1)$, the above equation takes the following form

$$\zeta_{i,j}^{n+1} = \zeta_{i,j}^{n} + \Delta t\left[-\frac{\Psi_{i,j+1}^{n} - \Psi_{i,j-1}^{n}}{2\Delta y}\frac{\zeta_{i+1,j}^{n} - \zeta_{i-1,j}^{n}}{2\Delta x} + \frac{\Psi_{i+1,j}^{n} - \Psi_{i-1,j}^{n}}{2\Delta x}\frac{\zeta_{i,j+1}^{n} - \zeta_{i,j-1}^{n}}{2\Delta y} + \frac{1}{Re}\left(\frac{\zeta_{i+1,j}^{n} - 2\zeta_{i,j}^{n} + \zeta_{i-1,j}^{n}}{\Delta x^2} + \frac{\zeta_{i,j+1}^{n} - 2\zeta_{i,j}^{n} + \zeta_{i,j-1}^{n}}{\Delta y^2}\right)\right] \tag{2.1.2}$$

In a similar way, the stream function equation can be expressed as:

$$\frac{\Psi_{i+1,j}^{n} - 2\Psi_{i,j}^{n} + \Psi_{i-1,j}^{n}}{\Delta x^2} + \frac{\Psi_{i,j+1}^{n} - 2\Psi_{i,j}^{n} + \Psi_{i,j-1}^{n}}{\Delta y^2} = -\zeta_{i,j}^{n} \tag{2.1.3}$$

Solving for $\Psi_{i,j}^{n}$ yields:

$$\Psi_{i,j}^{n} = \frac{\zeta_{i,j}^{n}(\Delta x)^2(\Delta y)^2 + (\Delta y)^2\,(\Psi_{i+1,j}^{n} + \Psi_{i-1,j}^{n}) + (\Delta x)^2(\Psi_{i,j+1}^{n} + \Psi_{i,j-1}^{n})}{2((\Delta x)^2 + (\Delta y)^2)} \tag{2.1.4}$$

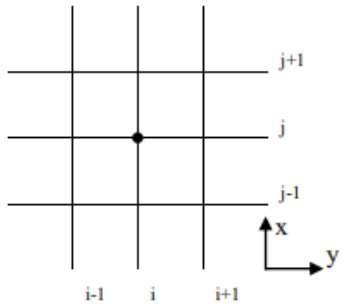

Figure 2.1.1. *Node layout schematic*

Solving equation (2.1.4) for $\Psi_{i,j}$ at the next iteration $(n+1)$ yields

$$\Psi_{i,j}^{n+1} = \frac{\zeta_{i,j}^{n}(\Delta x)^2(\Delta y)^2 + (\Delta y)^2\,(\Psi_{i+1,j}^{n} + \Psi_{i-1,j}^{n}) + (\Delta x)^2(\Psi_{i,j+1}^{n} + \Psi_{i,j-1}^{n})}{2((\Delta x)^2 + (\Delta y)^2)} \tag{2.1.5}$$

For equations (2.1.4) and (2.1.5), the superscripts n and $n+1$ do not represent time steps, as this would not be appropriate for solving elliptic equations. Instead, these values indicate the iteration step.

Relaxation methods are iterative techniques used for solving partial differential equations (PDEs), particularly elliptic PDEs. A relaxation technique involves iteratively updating the solution until convergence is achieved. The following describes the application of this method to equation (2.1.5).

Initially, Ψ is interpreted as an intermediate value, denoted by $\Psi_{i,j}^{n\bar{+}1}$, where

$$\Psi_{i,j}^{n\bar{+}1} = \frac{\zeta_{i,j}^{n}(\Delta x)^2(\Delta y)^2 + (\Delta y)^2(\Psi_{i+1,j}^{n} + \Psi_{i-1,j}^{n+1}) + (\Delta x)^2(\Psi_{i,j+1}^{n} + \Psi_{i,j-1}^{n+1})}{2((\Delta x)^2 + (\Delta y)^2)} \tag{2.1.6}$$

At this point, it is important to note that for the term $\Psi_{i-1,j}^{n+1}$, the iteration level $(n+1)$ is chosen, assuming that the calculations proceed from left to right, so this value is considered known. Similarly, for the term $\Psi_{i,j-1}^{n+1}$, the iteration level $(n+1)$ is used, assuming that the calculations proceed from bottom to top, so this value is also considered known.

Finally, we use the value $\Psi_{i,j}^{n}$ from the previous iteration together with $\Psi_{i,j}^{n\bar{+}1}$ from (2.1.6) to compute $\Psi_{i,j}^{n+1}$ via extrapolation

$$\Psi_{i,j}^{n+1} = \Psi_{i,j}^{n} + \omega\left(\Psi_{i,j}^{n\bar{+}1} - \Psi_{i,j}^{n}\right) \tag{2.1.7}$$

In equation (2.1.7), ω is the relaxation coefficient, whose value is usually determined by trial and error. If ω is greater than one, over-relaxation occurs; if ω is less than one, under-relaxation occurs.

2.2. Implicit Method

In the context of Computational Fluid Dynamics (CFD), an implicit method is a numerical approach used to solve the governing equations of fluid flow that involves solving a system of algebraic equations at each time step. In contrast to explicit methods, implicit methods do not update the solution directly from the known information at the current time step. Instead, they involve solving a set of equations typically arising from the discretization of the governing equations.

The basic idea behind these methods is to discretize the spatial and time domains and formulate a system of algebraic equations, often in the form of a linear or nonlinear system. This system is then solved at each time step, and the solution is updated accordingly. The solution of the system can be obtained using iterative techniques, depending on the nature of the problem and the characteristics of the discretization scheme.

The Alternating Direction Implicit (ADI) method is a numerical technique used for solving partial differential equations (PDEs), particularly parabolic-type equations. ADI is an implicit method that decomposes the solution process into two or more steps in alternating directions. This decomposition simplifies the numerical solution of multidimensional PDEs by transforming a multidimensional problem into a sequence of one-dimensional problems. ADI can provide improved numerical stability and efficiency compared to explicit methods.

For the specific problem, the domain is discretized into two-time steps. In the first-time step, from n to $\left(n + \frac{1}{2}\right)$, the x-direction is treated in implicit form, and in the next step, from n to $(n + 1)$, the y-direction is treated in implicit form. Using this intermediate step simplifies the equation, as it reduces the number of unknowns from five. The goal is to create two tridiagonal systems to determine the desired quantities.

For the step $n \rightarrow \left(n + \frac{1}{2}\right)$, the terms of equation (1.1) are discretized in the x-direction in implicit form. This yields

$$\frac{\zeta_{i,j}^{n+\frac{1}{2}}-\zeta_{i,j}^{n}}{\Delta t}=-\frac{\Psi_{i,j+1}^{n}-\Psi_{i,j-1}^{n}}{2\Delta y}\cdot\frac{\zeta_{i+1,j}^{n+\frac{1}{2}}-\zeta_{i-1,j}^{n+\frac{1}{2}}}{2\Delta x}+\frac{\Psi_{i+1,j}^{n}-\Psi_{i-1,j}^{n}}{2\Delta x}\cdot\frac{\zeta_{i,j+1}^{n}-\zeta_{i,j-1}^{n}}{2\Delta y}$$
$$+\frac{1}{Re}\left(\frac{\zeta_{i+1,j}^{n+\frac{1}{2}}-2\zeta_{i,j}^{n+\frac{1}{2}}+\zeta_{i-1,j}^{n+\frac{1}{2}}}{\Delta x^2}+\frac{\zeta_{i,j+1}^{n}-2\zeta_{i,j}^{n}+\zeta_{i,j-1}^{n}}{\Delta y^2}\right) \tag{2.2.1}$$

Rearranging the terms with $\left(n+\frac{1}{2}\right)$ on the left-hand side gives

$$\zeta_{i,j}^{n+\frac{1}{2}}\left(1+\frac{\Delta t}{Re\,\Delta x^2}\right)+\zeta_{i+1,j}^{n+\frac{1}{2}}\left(\frac{\Psi_{i,j+1}^{n}-\Psi_{i,j-1}^{n}}{2\Delta y}\cdot\frac{\Delta t}{4\Delta x}-\frac{\Delta t}{2Re\,\Delta x^2}\right)$$
$$+\zeta_{i-1,j}^{n+\frac{1}{2}}\left(-\frac{\Psi_{i,j+1}^{n}-\Psi_{i,j-1}^{n}}{2\Delta y}\cdot\frac{\Delta t}{4\Delta x}-\frac{\Delta t}{2Re\,\Delta x^2}\right)$$
$$=\zeta_{i,j}^{n}+\frac{\Delta t}{2}\left[\frac{\Psi_{i+1,j}^{n}-\Psi_{i-1,j}^{n}}{2\Delta x}\cdot\frac{\zeta_{i,j+1}^{n}-\zeta_{i,j-1}^{n}}{2\Delta y}+\frac{1}{Re}\frac{\zeta_{i,j+1}^{n}-2\zeta_{i,j}^{n}+\zeta_{i,j-1}^{n}}{\Delta y^2}\right] \tag{2.2.2}$$

The left-hand side contains the unknowns, while the right-hand side contains known values from the previous iteration.

For the step $\left(n+\frac{1}{2}\right)\rightarrow(n+1)$, the terms of equation (1.1) are discretized in the y-direction in implicit form, yielding

$$\frac{\zeta_{i,j}^{n+1}-\zeta_{i,j}^{n+\frac{1}{2}}}{\Delta t}=-\frac{\Psi_{i,j+1}^{n}-\Psi_{i,j-1}^{n}}{2\Delta y}\cdot\frac{\zeta_{i+1,j}^{n+\frac{1}{2}}-\zeta_{i-1,j}^{n+\frac{1}{2}}}{2\Delta x}+\frac{\Psi_{i+1,j}^{n}-\Psi_{i-1,j}^{n}}{2\Delta x}\cdot\frac{\zeta_{i,j+1}^{n+1}-\zeta_{i,j-1}^{n+1}}{2\Delta y}$$
$$+\frac{1}{Re}\left(\frac{\zeta_{i+1,j}^{n+\frac{1}{2}}-2\zeta_{i,j}^{n+\frac{1}{2}}+\zeta_{i-1,j}^{n+\frac{1}{2}}}{\Delta x^2}+\frac{\zeta_{i,j+1}^{n+1}-2\zeta_{i,j}^{n+1}+\zeta_{i,j-1}^{n+1}}{\Delta y^2}\right) \tag{2.2.3}$$

Rearranging terms with $n+1$ on the left-hand side gives

$$\zeta_{i,j}^{n+1}\left(1+\frac{\Delta t}{Re\,\Delta y^2}\right)+\zeta_{i,j+1}^{n+1}\left(\frac{\Psi_{i+1,j}^{n}-\Psi_{i-1,j}^{n}}{2\Delta x}\cdot\frac{\Delta t}{4\Delta y}-\frac{\Delta t}{2Re\,\Delta y^2}\right)$$
$$+\zeta_{i,j-1}^{n+1}\left(-\frac{\Psi_{i+1,j}^{n}-\Psi_{i-1,j}^{n}}{2\Delta x}\cdot\frac{\Delta t}{4\Delta y}-\frac{\Delta t}{2Re\,\Delta y^2}\right)$$
$$=\zeta_{i,j}^{n+\frac{1}{2}}+\frac{\Delta t}{2}\left[-\frac{\Psi_{i,j+1}^{n}-\Psi_{i,j-1}^{n}}{2\Delta y}\cdot\frac{\zeta_{i+1,j}^{n+\frac{1}{2}}-\zeta_{i-1,j}^{n+\frac{1}{2}}}{2\Delta x}+\frac{1}{Re}\frac{\zeta_{i+1,j}^{n+\frac{1}{2}}-2\zeta_{i,j}^{n+\frac{1}{2}}+\zeta_{i-1,j}^{n+\frac{1}{2}}}{\Delta x^2}\right] \tag{2.2.4}$$

where

$$u_{i,j}^{n} = \frac{\Psi_{i,j+1}^{n} - \Psi_{i,j-1}^{n}}{2\Delta y}, v_{i,j}^{n} = -\frac{\Psi_{i+1,j}^{n} - \Psi_{i-1,j}^{n}}{2\Delta x}$$

Equations (2.2.2) and (2.2.4) now form two tridiagonal systems, allowing the vorticity variables to be computed for the two-time steps. Specifically, using the Thomas Algorithm,

$$\zeta_{i-1,j}^{n+\frac{1}{2}}, \zeta_{i,j}^{n+\frac{1}{2}}, \zeta_{i+1,j}^{n+\frac{1}{2}}, \zeta_{i,j-1}^{n+1}, \zeta_{i,j}^{n+1}, \zeta_{i,j+1}^{n+1}$$

Similarly, for the step $n \to \left(n + \frac{1}{2}\right)$, the terms of equation (1.2) are discretized in the x-direction in implicit form, yielding

$$\frac{\Psi_{i+1,j}^{n+\frac{1}{2}} - 2\Psi_{i,j}^{n+\frac{1}{2}} + \Psi_{i-1,j}^{n+\frac{1}{2}}}{\Delta x^2} + \frac{\Psi_{i,j+1}^{n} - 2\Psi_{i,j}^{n} + \Psi_{i,j-1}^{n}}{\Delta y^2} = -\zeta_{i,j}^{n} \tag{2.2.5}$$

Rewriting with $\left(n + \frac{1}{2}\right)$ terms on the left gives

$$\Psi_{i+1,j}^{n+\frac{1}{2}} - 2\Psi_{i,j}^{n+\frac{1}{2}} + \Psi_{i-1,j}^{n+\frac{1}{2}} = -\zeta_{i,j}^{n}(\Delta x)^2 - \frac{\Psi_{i,j+1}^{n} - 2\Psi_{i,j}^{n} + \Psi_{i,j-1}^{n}}{\Delta y^2}(\Delta x)^2 \tag{2.2.6}$$

For the step $n + \frac{1}{2} \to n + 1$, the terms of equation (1.2) are discretized in the y-direction in implicit form, yielding

$$\frac{\Psi_{i+1,j}^{n+\frac{1}{2}} - 2\Psi_{i,j}^{n+\frac{1}{2}} + \Psi_{i-1,j}^{n+\frac{1}{2}}}{\Delta x^2} + \frac{\Psi_{i,j+1}^{n+1} - 2\Psi_{i,j}^{n+1} + \Psi_{i,j-1}^{n+1}}{\Delta y^2} = -\zeta_{i,j}^{n} \tag{2.2.7}$$

Rewriting with $(n + 1)$ terms on the left gives

$$\Psi_{i,j+1}^{n+1} - 2\Psi_{i,j}^{n+1} + \Psi_{i,j-1}^{n+1} = -\zeta_{i,j}^{n}(\Delta y)^2 - \frac{\Psi_{i+1,j}^{n+\frac{1}{2}} - 2\Psi_{i,j}^{n+\frac{1}{2}} + \Psi_{i-1,j}^{n+\frac{1}{2}}}{\Delta x^2}(\Delta y)^2 \tag{2.2.8}$$

Equations (2.2.6) and (2.2.8) now form two tridiagonal systems, allowing the streamfunction variables to be computed for the two-time steps. Specifically, using the Thomas Algorithm,

$$\Psi_{i-1,j}^{n+\frac{1}{2}}, \Psi_{i,j}^{n+\frac{1}{2}}, \Psi_{i+1,j}^{n+\frac{1}{2}}, \Psi_{i,j-1}^{n+1}, \Psi_{i,j}^{n+1}, \Psi_{i,j+1}^{n+1}.$$

2.3. Boundary Conditions

The boundary conditions for vorticity ζ and the streamfunction Ψ are set so that:

- Velocities $u = U$ and $v = 0$ for the top wall.
- Velocities $u = v = 0$ for the other three walls.

For a stationary wall, we have

$$v = 0 \Rightarrow \frac{\partial v}{\partial x} = 0$$

and therefore, the vorticity takes the form

$$\zeta = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = -\frac{\partial^2 \Psi}{\partial y^2}$$

where

$$\frac{\partial \Psi}{\partial y} = u, \frac{\partial \Psi}{\partial x} = -v$$

Using a Taylor expansion of the streamfunction at the bottom wall, for $v = 0 = \frac{\partial \Psi}{\partial y}$, we have

$$\Psi_{i,2} = \Psi_{i,1} + \left(\frac{\partial \Psi}{\partial y}\right)_{i,1} \Delta y + \frac{1}{2}\left(\frac{\partial^2 \Psi}{\partial y^2}\right)_{i,1} (\Delta y)^2 \Rightarrow$$

$$\Psi_{i,2} = \Psi_{i,1} + 0 + \frac{1}{2}(-\zeta_{i,1})(\Delta y)^2 \Rightarrow \zeta_{i,1} = \frac{2(\Psi_{i,1} - \Psi_{i,2})}{(\Delta y)^2} \tag{2.3.1}$$

Similarly, for the left wall

$$\Psi_{2,j} = \Psi_{1,j} + \left(\frac{\partial \Psi}{\partial x}\right)_{1,j} \Delta x + \frac{1}{2}\left(\frac{\partial^2 \Psi}{\partial x^2}\right)_{1,j} (\Delta x)^2 \Rightarrow$$

$$\Psi_{2,j} = \Psi_{1,j} + 0 + \frac{1}{2}(-\zeta_{1,j})(\Delta x)^2 \Rightarrow \zeta_{1,j} = \frac{2(\Psi_{1,j} - \Psi_{2,j})}{(\Delta x)^2} \tag{2.3.2}$$

For the right wall

$$\Psi_{end-1,j} = \Psi_{end,j} + \left(\frac{\partial \Psi}{\partial x}\right)_{end,j} \Delta x + \frac{1}{2}\left(\frac{\partial^2 \Psi}{\partial x^2}\right)_{end,j} (\Delta x)^2 \Rightarrow$$

$$\Psi_{end-1,j} = \Psi_{end,j} + 0 + \frac{1}{2}\left(-\zeta_{end,j}\right)(\Delta x)^2 \Rightarrow \zeta_{end,j} = \frac{2\left(\Psi_{end,j} - \Psi_{end-1,j}\right)}{(\Delta x)^2} \qquad (2.3.3)$$

Finally, for the top wall, where $\frac{\partial \Psi}{\partial y} = u = U$, and since $\zeta = -\frac{\partial^2 \Psi}{\partial y^2}$, a Taylor expansion gives:

$$\Psi_{i,end-1} = \Psi_{i,end} + \left(\frac{\partial \Psi}{\partial y}\right)_{i,end} \Delta y + \frac{1}{2}\left(\frac{\partial^2 \Psi}{\partial y^2}\right)_{i,end} (\Delta y)^2 \Rightarrow \zeta_{i,end} = \frac{2\left(\Psi_{i,end} - \Psi_{i,end-1}\right) - 2U\Delta y}{(\Delta y)^2} \qquad (2.3.4)$$

3. Results

The objective of this section is to compute the flow field using both the explicit and implicit formulations, as well as to plot the streamlines and vorticity contours for three Reynolds numbers: Re = 1, 100, and 500. The velocity at the upper wall is set to 1 m/s and remains constant.

The relative error was defined as $|\ \varepsilon_{n+1} - \varepsilon_n\ |$. For the convergence of the iterative process, that is, when the solution reaches a steady state, a relative error smaller than 10^{-4} was chosen.

3.1. Re=1

For a Reynolds number equal to 100, the parameters used were $\Delta x = \Delta y = 0.01$, corresponding to a 101×101 grid, and $\Delta t = 0.0001$. The relaxation factor was set to $\omega = 0.9$.

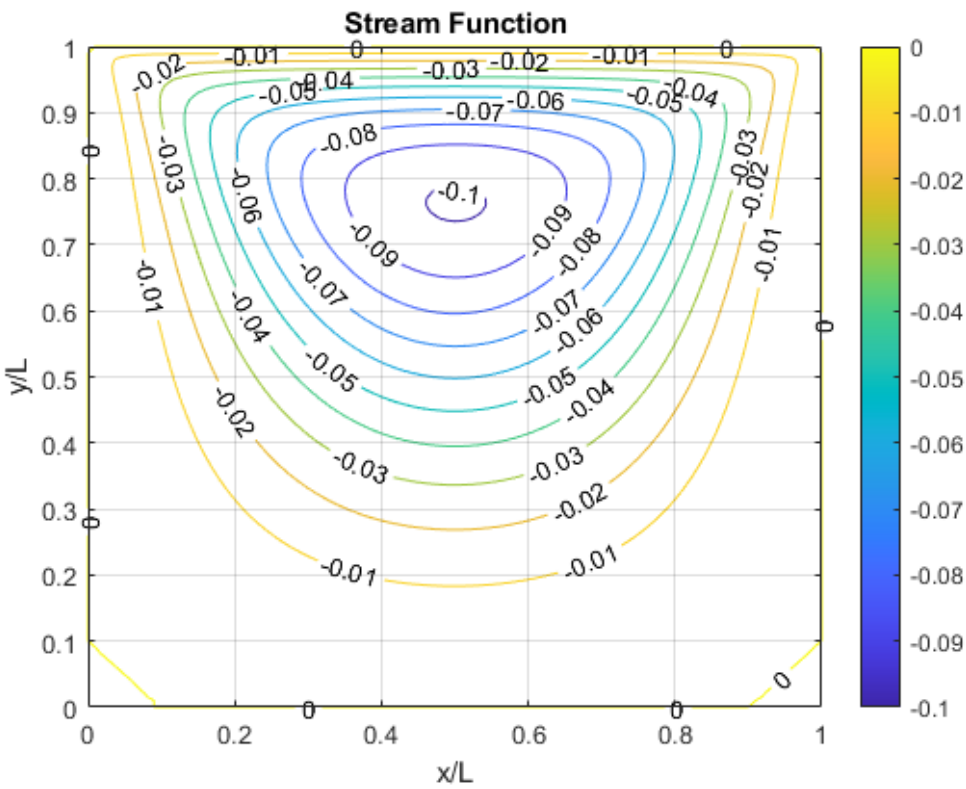

Figure 3.1.1. *Contour plot of the stream function for Re = 1*

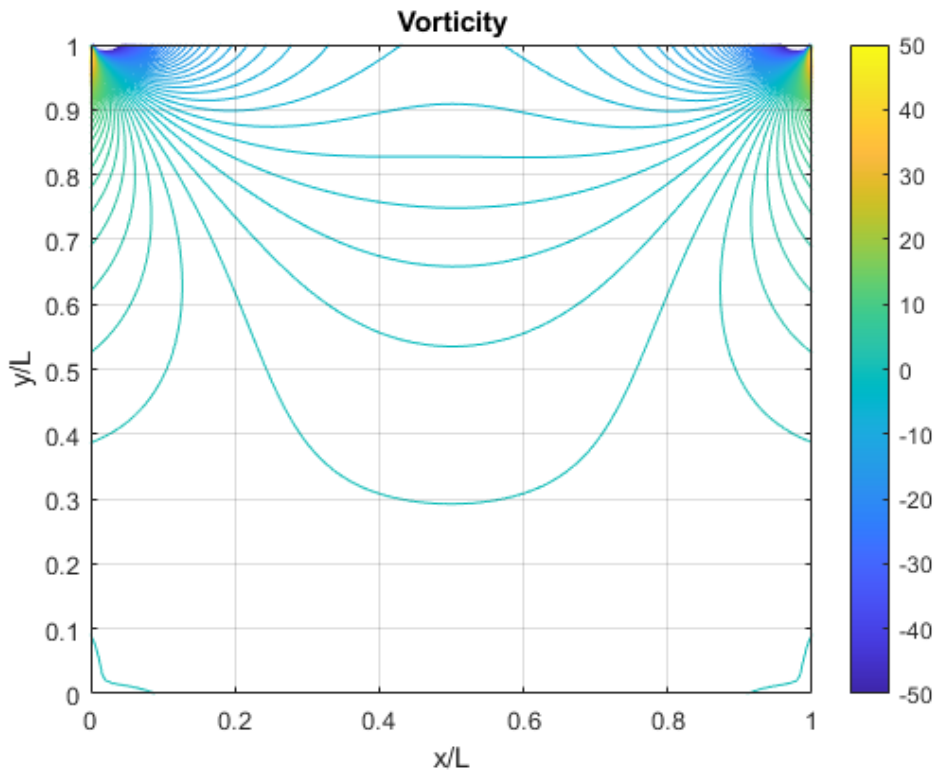

Figure 3.1.2. *Contour plot of vorticity for Re=1*

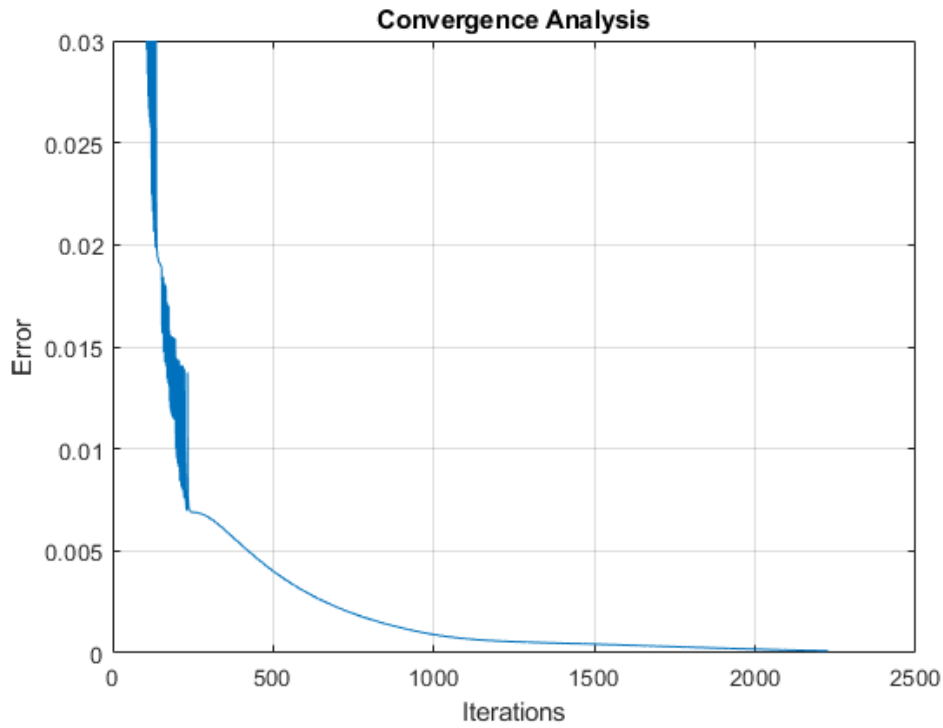

Figure 3.1.3. *Convergence analysis for Re=1*

For a Reynolds number equal to 1, as shown in Figures 3.1.1 and 3.1.2, there is once again a good agreement between the reference plots and those obtained from the computational code. Furthermore, it is observed that the highest vorticity values are, correctly, located near the walls. Finally, convergence is achieved after approximately 1500-2000 iterations, where the error becomes smaller than 10^{-4}.

3.2. Re=100

For a Reynolds number of 100, $\Delta x = \Delta y = 0.01$ was used, corresponding to a 101×101 grid, along with $\Delta t = 0.0001$. The relaxation factor was chosen as $\omega = 0.9$.

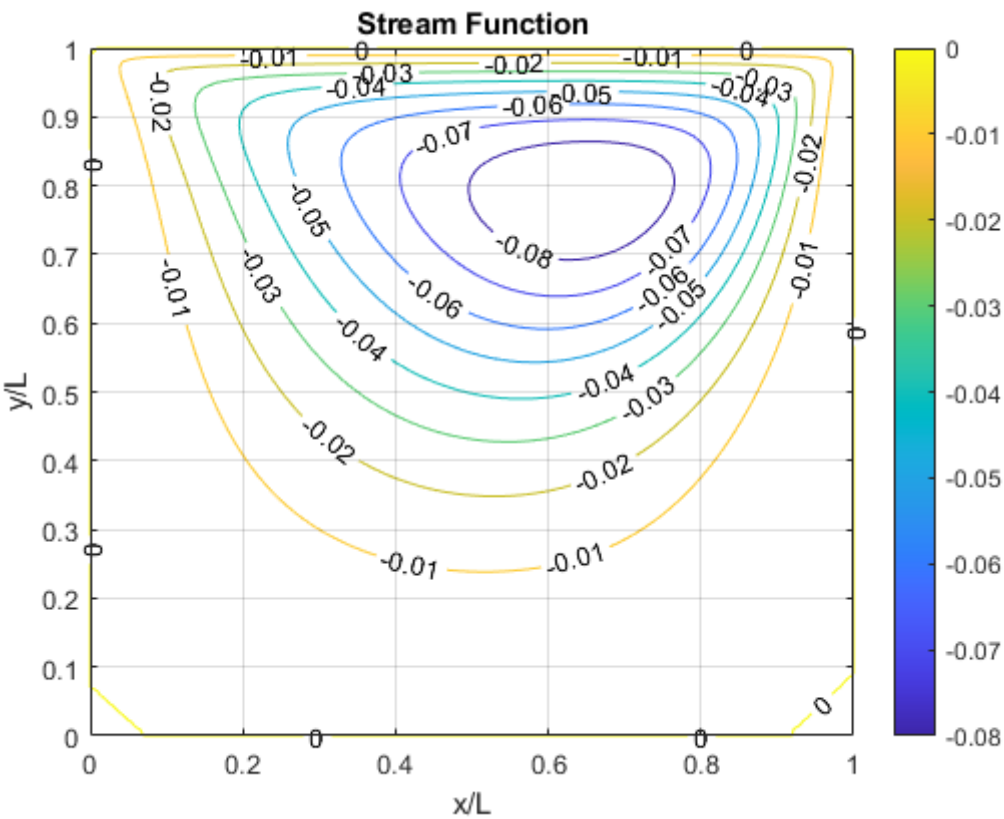

Figure 3.2.1: *Contour plot of the stream function for Re = 100*

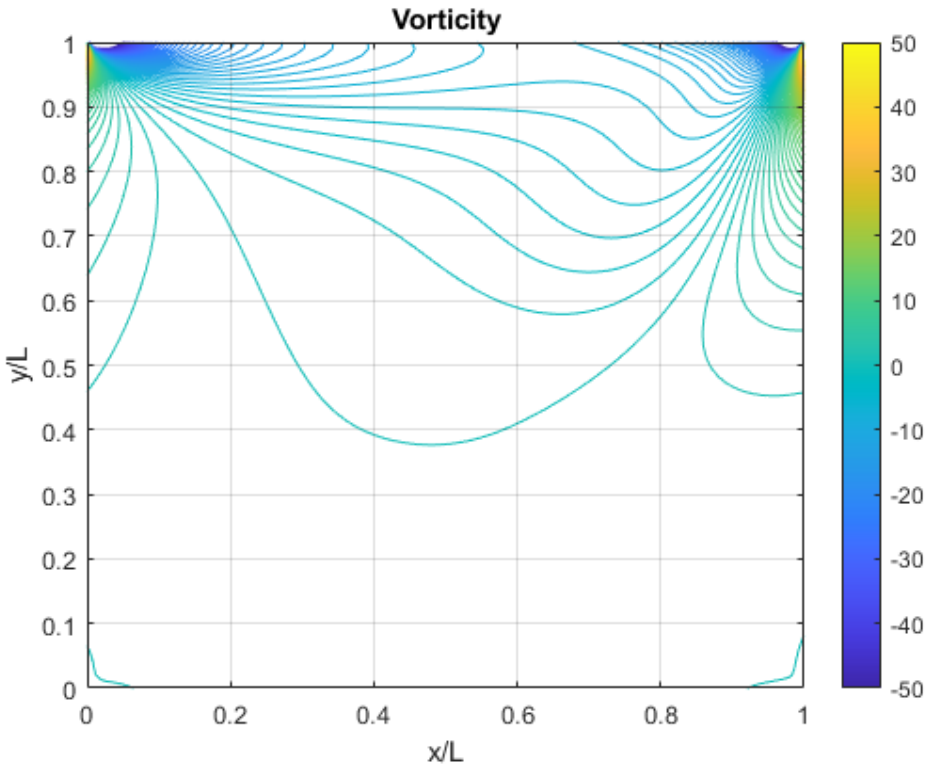

Figure 3.2.2. *Contour plot of vorticity for Re=100*

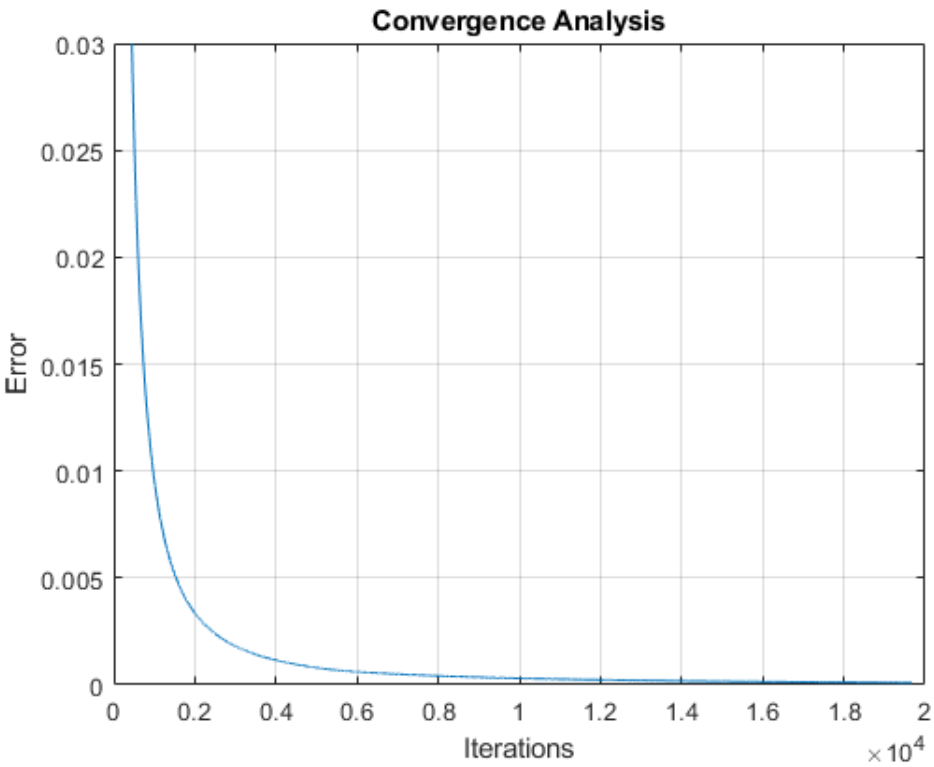

Figure 3.2.3. *Convergence analysis for Re=100*

For a Reynolds number of 100, according to Figures 3.2.1 and 3.2.2, there is good agreement between the reference plots and those obtained from the code. A shift of the streamlines and vorticity toward the right is observed compared to Re = 1, consistent with expectations. Furthermore, convergence is achieved after approximately 10000-12000 iterations, with the error falling below 10^{-4}.

3.3. Re=500

For a Reynolds number equal to 500, $\Delta x = \Delta y = 0.01$ was used, corresponding to a 101×101 grid, along with $\Delta t = 0.0001$. The relaxation factor was chosen as $\omega = 0.9$.

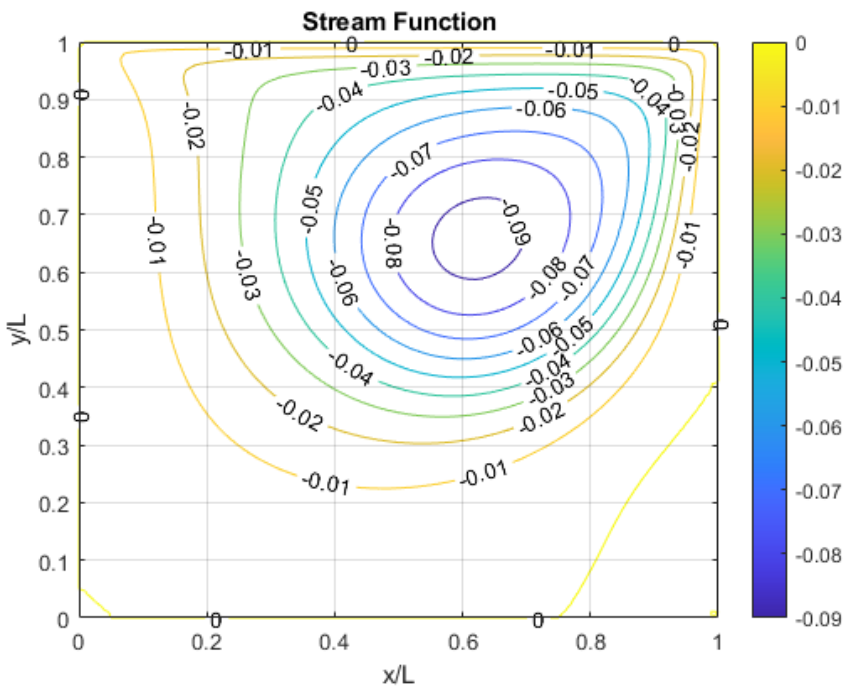

Figure 3.3.1. *Contour plot of the stream function for Re = 500*

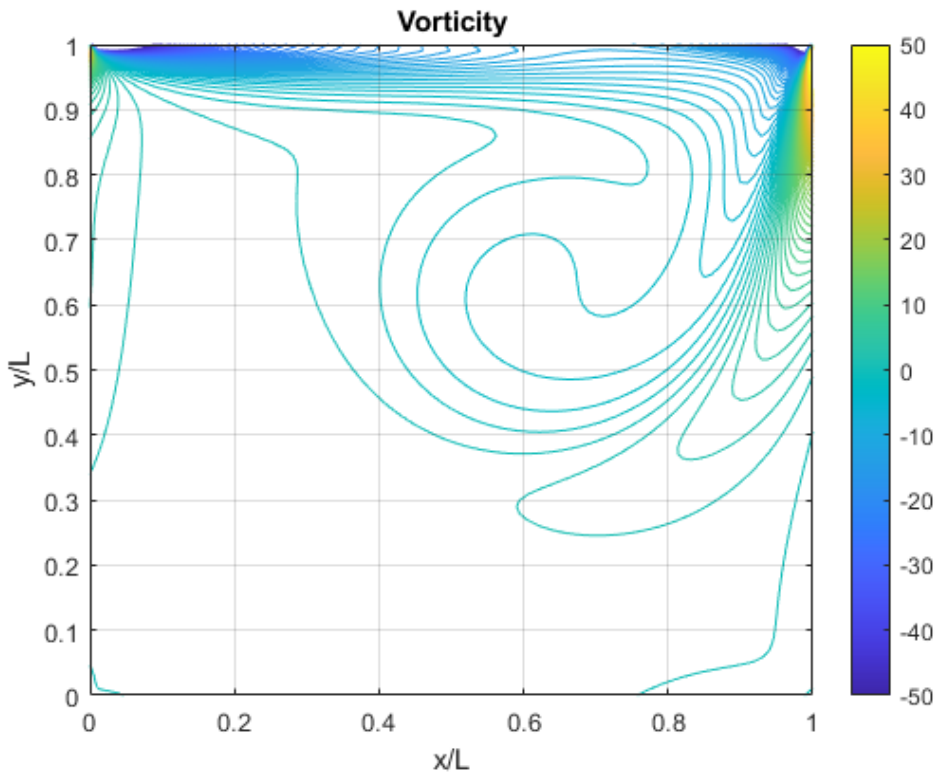

Figure 3.3.2. *Contour plot of vorticity for Re=500*

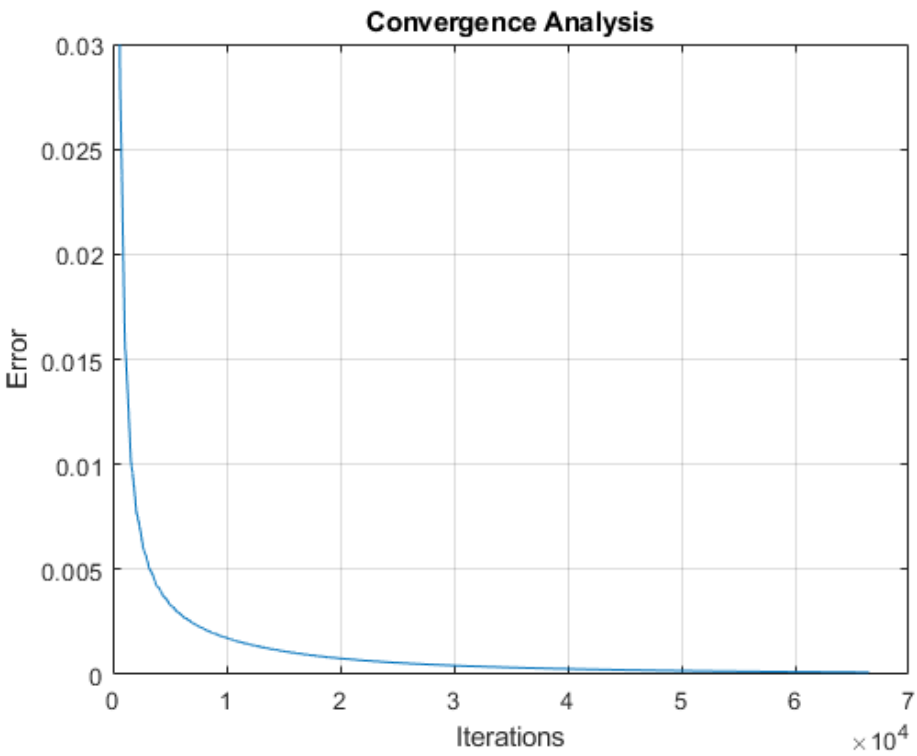

Figure 3.3.3. *Convergence analysis for Re=500*

For Reynolds number equal to 500, according to Figures 3.3.1 and 3.3.2, there is a shift of the streamlines to the right, along with a stronger vorticity phenomenon. Finally, convergence is achieved after approximately 4000–5000 iterations, with the error becoming smaller than 10^{-4}.

4. Grid Independence Study

For the grid independence study, four types of grids were selected for comparison. Specifically:

- Grid 1: 125×125 → ($\Delta x = \Delta y = 0.008$)
- Grid 2: 143×143 → ($\Delta x = \Delta y = 0.007$)
- Grid 3: 166×166 → ($\Delta x = \Delta y = 0.006$)
- Grid 4: 101×101 → ($\Delta x = \Delta y = 0.01$)

A constant Reynolds number (Re = 100) was chosen to ensure meaningful comparisons, while the time step was also kept constant at $\Delta t = 0.0001$.

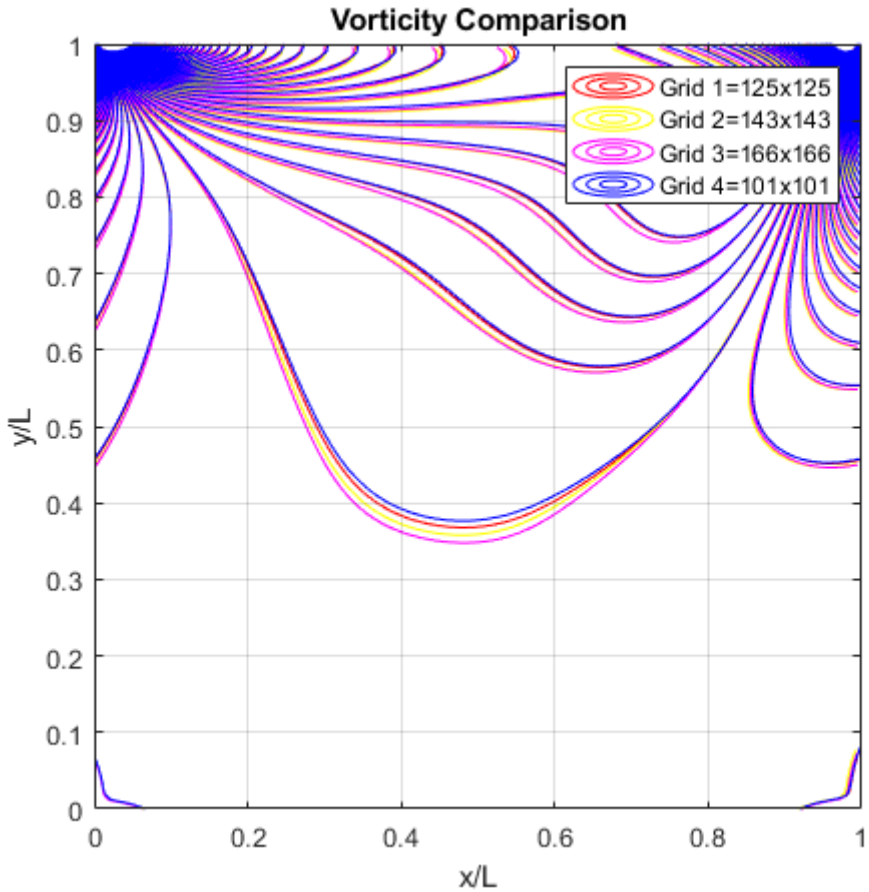

Figure 4.1. *Comparison of vorticity contour plots*

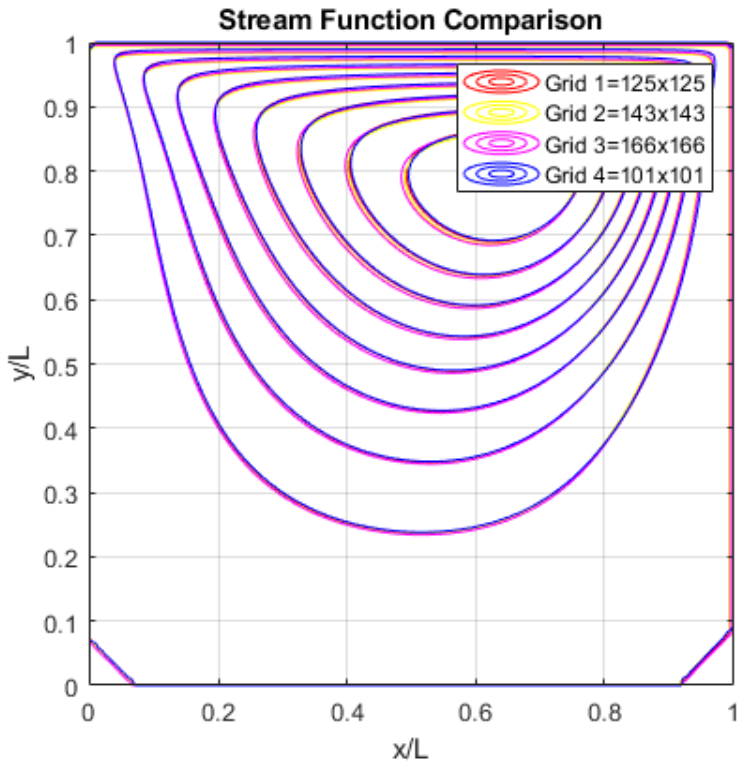

Figure 4.2. *Comparison of streamfunction contour plots*

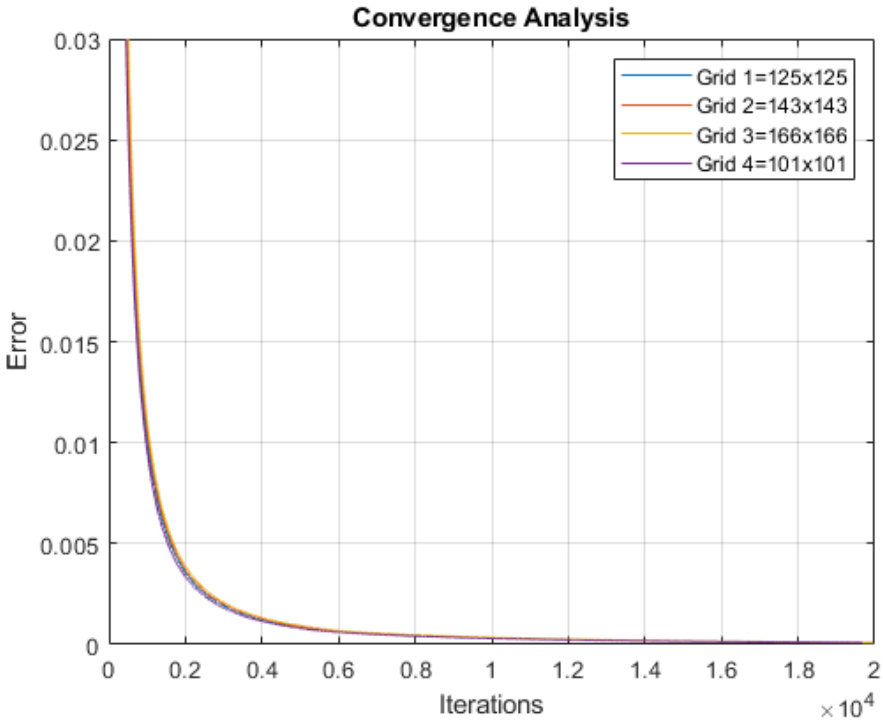

Figure 4.3. *Convergence analysis*

From the above plots, a good agreement of values across different grids can be observed. As the grid becomes finer, no significant difference is noted in either the stream function or the vorticity. On the other hand, an increase in computational time was observed, which did not substantially improve the final solution. Specifically, the required computation times for the grids were 28, 35, and 47 seconds, compared to 20 seconds for the initial grid. Finally, in all cases, the relative convergence error follows a similar pattern and reaches the threshold of 10^{-4} after approximately the same number of iterations.

5. Opposite Direction of Flow

The objective of this chapter is to study the same problem, with the only difference being that the velocity of the upper wall is reversed, that is, directed from right to left. A Reynolds number of 100 was used, with $\Delta x = \Delta y = 0.01$ (resulting in a 101×101 grid) and a time step of $\Delta t = 0.0001$. The relaxation factor was set to $\omega = 0.9$. The same parameters as in the previous section were chosen to enable a direct and more accurate comparison.

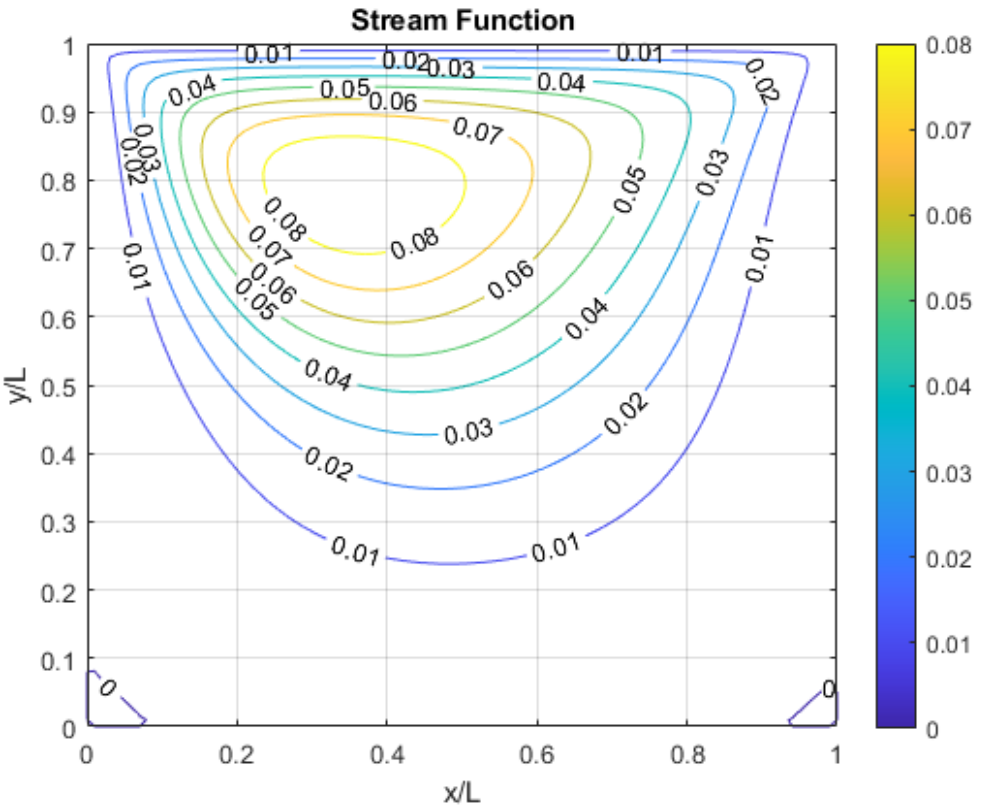

Figure 5.1. *Contour plot of the streamfunction for Re = 100*

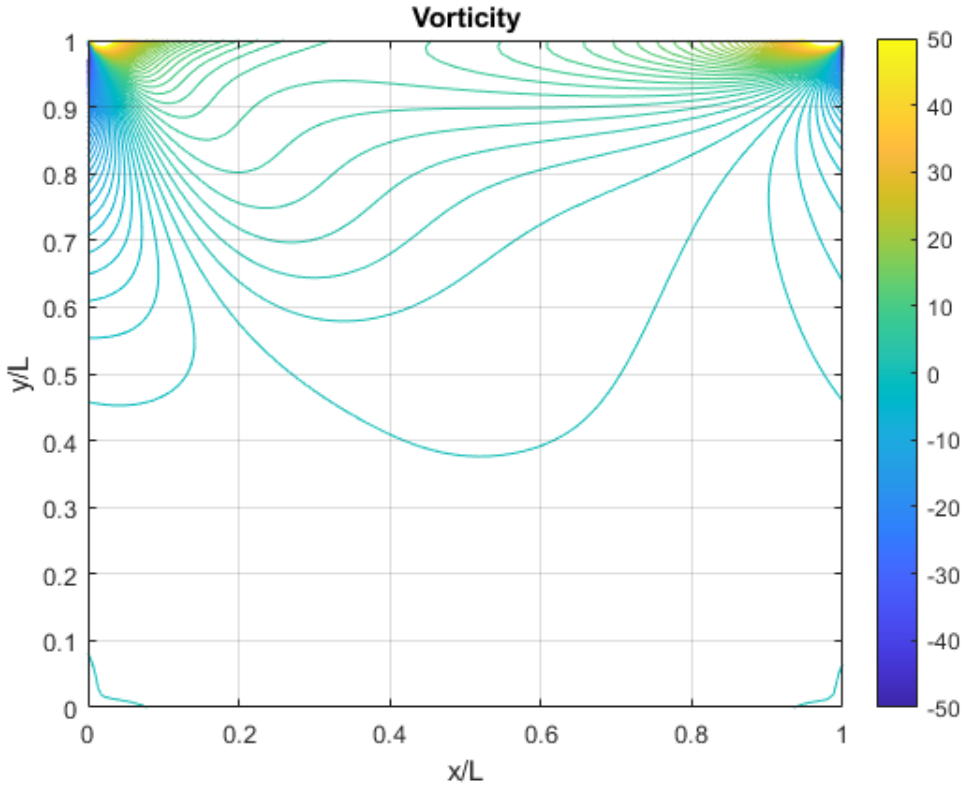

Figure 5.2. *Contour plot of vorticity for Re=100*

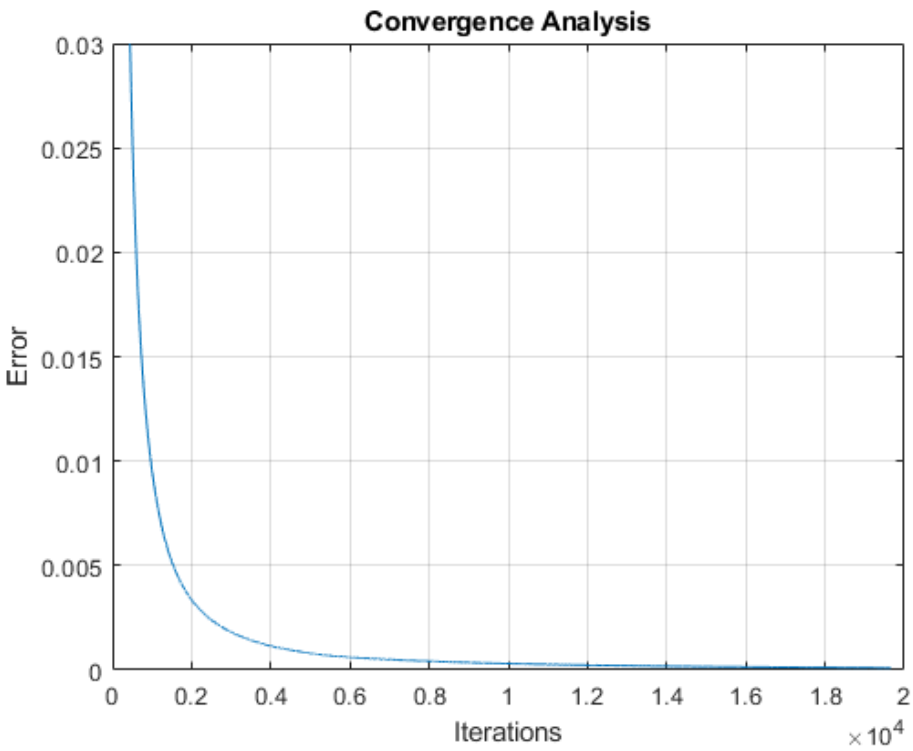

Figure 5.3. *Convergence analysis for Re=100*

As expected, the results are similar to those obtained when the plate velocity is positive. Specifically, the streamlines exhibit the same behavior, while the vortex is symmetric but oriented in the opposite direction. Finally, the error convergence follows the same pattern and reaches the same level after an equivalent number of iterations.